

RESEARCH ARTICLE

A Web-Based Interface That Leverages Machine Learning to Assess an Individual's Vulnerability to Brain Stroke

DIVYANSH BHANDARI¹, ARNAV AGARWAL¹, R. REENA ROY²,
RAJARAM PRIYATHARSHINI³, AND RODRIGUEZ RIVERO CRISTIAN^{4,5}

¹School of Electronics Engineering, Vellore Institute of Technology, Chennai, Tamil Nadu 600127, India

²School of Computer Science and Engineering, Vellore Institute of Technology, Chennai, Tamil Nadu 600127, India

³Centre for Engineering Research in Intelligent Sensors and Systems, Cardiff School of Technologies, Cardiff Metropolitan University, CF5 2YB Cardiff, U.K.

⁴Cardiff School of Technologies, Cardiff Metropolitan University, CF5 2YB Cardiff, U.K.

⁵Department of Mining, Industrial, and ICT Systems Engineering, Universitat Politècnica de Catalunya, 08034 Barcelona, Spain

Corresponding author: R. Reena Roy (reenaroy.r@vit.ac.in)

ABSTRACT Cerebral stroke is a major global health issue, contributing to high mortality and long-term disability. Early identification of individuals at high risk of stroke can significantly improve preventive care outcomes. We present a web-based stroke risk assessment tool that uniquely combines an accessible user interface with robust machine learning modeling. The proposed platform leverages a novel combination of SMOTE oversampling and logistic regression to address class imbalance in patient health records, improving the detection of stroke risk factors over existing methods. We compare a range of algorithms—including traditional classifiers and deep learning models—and report comprehensive performance metrics (accuracy, precision, recall, F1-score, and AUC-ROC) for each. Our best model (logistic regression with SMOTE and standard scaling) achieves 93.2% accuracy with a substantially higher F1-score for the stroke-positive class than other models, indicating improved sensitivity to stroke cases. To bridge the gap between complex predictive models and end-users, we deploy this model in an intuitive web interface (built with Streamlit) that non-technical individuals and healthcare providers can easily use. This interface requires no specialized knowledge, preserves user privacy by avoiding any data storage, and provides clear explanations of results. By offering a practical and transparent tool for stroke risk screening, our work advances health informatics with an emphasis on accessibility, interpretability, and early intervention. Potential applications range from personal health self-assessment to integration in clinical workflows for preventive care, ultimately aiming to improve public health outcomes through early detection and intervention in stroke.

INDEX TERMS Deep learning, early stroke detection, machine learning, predictive modeling, public health, stroke risk assessment, web-based interface.

I. INTRODUCTION

Stroke (cerebrovascular accident) occurs when blood supply to the brain is disrupted by blockage or hemorrhage, leading to brain tissue damage. It is among the leading causes of death and disability worldwide. In low- and middle-income regions alone, over 11 million strokes occur annually, causing about 4 million deaths and leaving roughly one-third of survivors

severely disabled [1]. This confirms the fact that in the case of early diagnosis and treatment, the gravity of outcomes and the patients' prognosis can be mitigated meaningfully. Stroke, also known as a cerebrovascular accident, is the impairment of a part of the brain due to its blood supply being disrupted. An interruption of the blood supply to the brain results in a stroke, which can be an ischemic or a hemorrhagic stroke [2]. Strokes can be broadly classified into two main types: While Ischemic stroke is a condition characterized by the narrowing or blockage in a brain's blood supply, Hemorrhagic stroke

The associate editor coordinating the review of this manuscript and approving it for publication was Muammar Muhammad Kabir¹.

happens when there is rupture of the blood vessels supplying the brain [3]. Since there is no cure for stroke, early detection can prevent deaths [4]. As a result, if stroke is diagnosed as soon as possible and appropriate treatment is administered, the mortality rate and resulting severe brain impairment would be minimized with success rate of up to 85% [5]. Pre-stroke symptoms, such as transient ischemic attacks (TIAs), which are brief disruptions in blood flow to the brain, can signal a high risk of stroke in the near future. Machine learning and deep learning models are increasingly used to predict stroke based on predisposing factors like BMI, smoking status, and others. However, the inherent complexity of some models can limit their accessibility for general public use, prompting research into methods like explainable AI to improve understanding and utilization of these predictive tools [6].

In response to this, we have created a web-based input/output platform, which uses machine learning to assess an individual's risk of experiencing a brain stroke. This interface is best used in advance of the CVD records check to allow people to gauge the basic level of risk they may be in. It is based on a model trained on the data from Kaggle, which offered the best test performance among 11 basic ML models and 5 DL models, as well as such variations as the grid-searched models of the basic ML models, and 3 types of scalers and 3 types of oversamplers for all the models mentioned above. The existing stroke prediction tools primarily utilize Logistic Regression, Decision Trees, and Random Forests, often not performing up to the expectations in classifying stroke risk specifically with imbalanced health data. Our approach incorporates advanced SMOTE oversampling with Standard Scaler and Logistic Regression, achieving a 93.2% accuracy in stroke prediction, which demonstrates significant improvement over commonly used models. Furthermore, the tool's web-based design enhances accessibility and ease of use for both patients and healthcare providers, bridging the gap between sophisticated predictive models and public accessibility in health informatics. By leveraging a user-centered design, this tool aims to extend stroke risk prediction to a broad audience, supporting public health initiatives in early intervention.

This paper introduces a novel web-based interface for stroke risk assessment, designed to be both technologically robust and accessible to users without technical expertise. Unlike prior tools that exist as research code or complex clinical software, our approach prioritizes ease of use and broad accessibility.

The key contributions of our work include:

- **User-Centered Web Interface:** We develop a publicly accessible web application using Streamlit, where individuals can input health information (e.g., age, BMI, blood pressure, lifestyle habits) and instantly receive a stroke risk evaluation. By resembling an online questionnaire, our tool eliminates the need for installation or specialist knowledge, making stroke risk screening widely available.

- **Enhanced Predictive Modeling for Imbalanced Data:** Our system employs a novel combination of Synthetic Minority Oversampling Technique (SMOTE) and logistic regression, along with standard feature scaling. This method improves stroke-prone case detection in an imbalanced dataset, achieving a better balance between sensitivity and specificity compared to baseline models.
- **Comprehensive Algorithm Benchmarking:** We evaluated 11 machine learning models and 5 deep learning architectures using multiple performance metrics, including precision, recall, F1-score, and AUC-ROC. While deep learning models like LSTMs and GRUs were tested, they were excluded from the final system due to the non-temporal nature of our dataset.
- **Transparency and Interpretability:** Unlike black-box models, our logistic regression model provides interpretable coefficients for each risk factor. This allows for the future integration of explainable AI techniques, such as SHAP values, to clarify how each feature contributes to stroke risk predictions—an aspect often overlooked in similar tools.
- **Validation and Reliability:** We incorporated rigorous validation methods, including cross-validation, hyperparameter tuning, and scenario stress-testing inspired by statistical model-checking techniques. This ensures that our model behaves consistently across various conditions, enhancing its reliability for real-world use.

Through this work, we aim to address existing gaps in stroke risk assessment by offering a scalable, user-friendly, and interpretable predictive tool. We hope this research contributes to the advancement of Health Informatics and encourages further exploration in this area.

Our study addresses the following research questions:

- 1) Which combinations of machine learning algorithms, oversampling methods (e.g., SMOTE), and scaling techniques yield the most accurate and reliable stroke risk predictions for imbalanced healthcare data?
- 2) How do traditional machine learning models compare to deep learning architectures in predicting stroke susceptibility from the same dataset, especially regarding class imbalance and interpretability?
- 3) To what extent does deploying a simple, user-friendly web-based interface improve public accessibility to stroke risk assessment tools, and does it facilitate earlier preventive action among non-technical users?

II. RELATED WORKS

Continuing from the works done previously, Shaily sought research how machine learning algorithm can help in stroke detection, which is a significant task because the disease has many dimensions and causes [7]. By making use of machine learning, Chang et al. studied facial stroke detection with reference to expressional asymmetry and mouth askew being some of the common facial traits of a stroke [8]. Kashi et al. in their study followed a fascinating approach where they used a machine learning system to identify the movement

compensations in stroke patients [9]. Other papers are related to the utilization of deep learning and machine learning as a method for identifying the early stage of stroke and hemorrhage by the researchers such as Al-Mekhlafi. Their work specifically involved conducting experiments where the AlexNet model and a fusion of deep learning–machine learning were employed for classifying cerebral hemorrhage using MRI data [10]. Malini et al. have designed a cloud-supported prediction system with the support of the machine learning algorithm that can monitor early possibilities of strokes. To ensure quick identification of a stroke and the subsequent notification of other personnel, this system focuses on the swift identification of the condition [11]. Tusher et al. showcased a look at a machine learning system which aims at predicting early onset stroke. This specifically applied the classification algorithms of Logistic Regression, C&RT, KNN, and SVM [12]. In the paper by Raja and others, machine learning algorithms enhancing stroke prognosis were examined. Their motivation was to make a sound health care service to detect the signs of stroke and had better probabilities to predict the same [13]. These papers on the state of stroke detection with the help of machine learning algorithms highlight several potential problems with the current status of the topic as follows: the unavailability of the model in clinical or at home environment, means that it requires manual modifications before use hence making it slow, inability to detect early signs of stroke using MRI data sets and the fact that it depends on few facial elements to achieve detection of the stroke thus implying that it might not be comprehensive enough to pick all the strokes. Concerns exist over the prompt notification of medical personnel or carers, as well as the general efficacy of predictive models in correctly identifying stroke cases.

The most common techniques in the detection of stroke are reading CT scan and MRI scan, as observed from other studies [14], [15], [16], [17], [18], [19], [20]. One of the ways through which the models can read the brain scans is because indeed the color differences are unique, that is, a different tissue in the brain is represented with a different hue [15]. Starting with diagnosis, continued through primary and secondary prevention, and finally to rehabilitation and treatment of stroke, the Markov Chain Monte Carlo-ML and DL algorithms seem to have considerable potential influence. There are algorithms which can both identify it and differentiate between the various types of stroke. While the performance metrics may vary significantly when different types of scans are compared, different models present varying levels of accuracy [14], [16], [17], [18], [19]. And as a result, we can state that no fixed algorithm can reach high-end outcomes each time it struggles with a scan. In these papers, we were able to classify scenarios based on the types of algorithms employed, the type of preprocessing applied and the nature of the datasets in finding the final solution. For this purpose, various 3D image reading algorithms are being employed to identify the best option and therefore the best solution [14], [20]. Recent advancements in model

verification, such as the use of OSM frameworks [21], provide rigorous checks on system behaviors, especially under dynamic conditions. In healthcare, this approach supports the reliable assessment of predictive models, ensuring their outputs align with clinical expectations. Our tool, though not leveraging OSM directly, incorporates similar verification principles at its core through thorough testing on the data. While several recent studies have explored deep learning, ensemble models, and explainable AI (XAI) for stroke prediction, our focus remains on developing an accessible and interpretable solution rather than prioritizing model complexity. Advanced architectures such as XGBoost, CNNs, and hybrid deep learning models have shown potential in improving predictive accuracy; however, they often lack the transparency required for healthcare applications. Given the importance of interpretability and usability for non-technical users, we prioritized traditional machine learning models, which offer well-understood decision-making processes that can be more easily adapted for real-world implementation. Future research could explore balancing complexity and interpretability to incorporate deep learning while maintaining clinical transparency.

III. METHODOLOGY

In this paper, we give an overview of our system, and find out that there are several important steps. Our proposed system contains several essential steps. Some of these are closer to basics, and they involve using data imputation or data splitting and also applying one-hot encoding on multiple features. Scaling applied to cleaning are utilized along with oversampling techniques to be fed into different combinations of ML and DL models. Evaluation of these strategies is done by comparing the F1-Score metric, but only for the minority class. This model is then selected as the best performing, and the next section involves developing an interactive web-based UI using Streamlit to build the interface.

The data used in this study was obtained from Kaggle [22] and a number of patients' characteristics defining their lifestyle was available, which can significantly influence the early development of stroke, including gender, age, BMI, smoking etc. Random checks made on the raw dataset revealed that the pre-processing had involved a total of 5110 observations with 12 attributes.

A. DATA PREPARATION

The first and most obvious step of records and transforming data set data cleaning process include was the deletion of the "ID" column. This was because the "ID" column had no corresponding data in other columns, and removing it prevented confusion during model training..

Next, we went over the data and made the last decisions of pre-processing before further analysis. In this regard, We removed one record classified as 'other' under gender. This was done so in a view to avoid time and effort wastage at the data preprocessing phase and because this entry in our

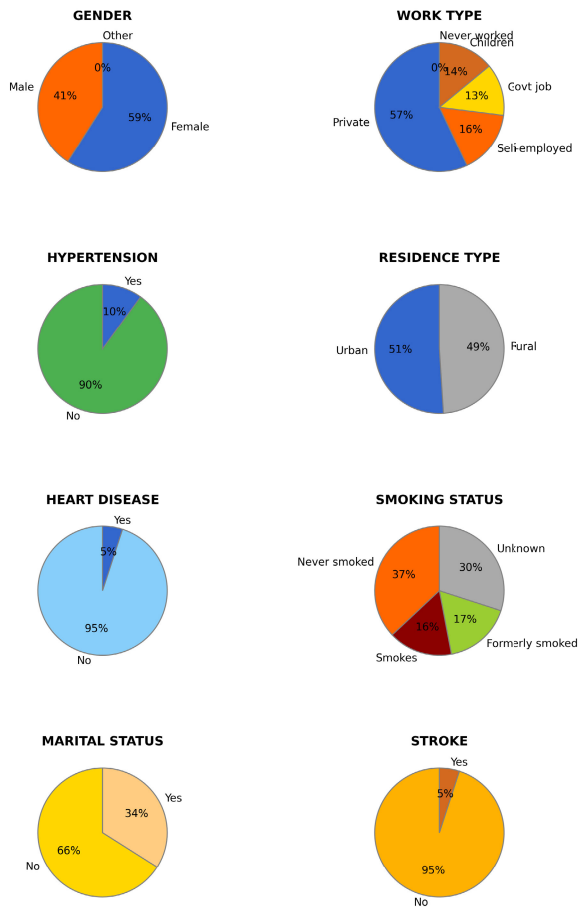


FIGURE 1. Distribution of features (a) gender (b) hypertension (c) heart disease (d) marital status (e) work type (f) residence type (g) smoke status (h) outcome of stroke, in the raw data.

TABLE 1. One-hot encoding for the listed attributes.

Attribute	Original	Changed
Gender	Male	1
	Female	0
Ever Married	Yes	1
	No	0
Residence Type	Urban	1
	Rural	0

case was unlikely to bring in a huge number of observations out of the total observation of 5110 into the dataset. Finally, we did some data preprocessing, where we transformed all the categorical data available in the dataset to numerical data. This was achieved using “label_encoder” from scikit-learn Python library. The labels, which were transformed for the given attributes, are listed in Table 1.

Proceeding from the data pre-processing, we improved the given dataset contrary to the categorical features transform “work_type” and “smoking_status” into the numerical format. This was done by following the example of the

“get_dummies” function of the pandas library in the Python language. When the data is categorical data in their raw form, applying machine learning models is challenging. One suggestion addresses this problem by handling these variables in a way that makes sense to computers, which improves the overall accuracy of the prediction. One-hot encoding is a technique that helps solve trouble that is common in machine learning: working with categorical values. The working algorithm of this encoding is to transform each given unique category into the respective binary feature. This means that a novel vector is created in which the value 1 reveals the existence of a particular category, while the absence of a category returns a 0. This has been important in making sure that the machine learning models that are used can handle the data properly. Every two vectors whose elements are either 0 or 1 are orthogonal to each other. This process enhances the ability of the categorical data representation to be more expressive to the algorithms understanding. Earlier, we have also discussed how it is possible to use the “get_dummies” method to produce dummy variables. This method provides for a new variable of categorical type into a set of variables that are binary in nature.

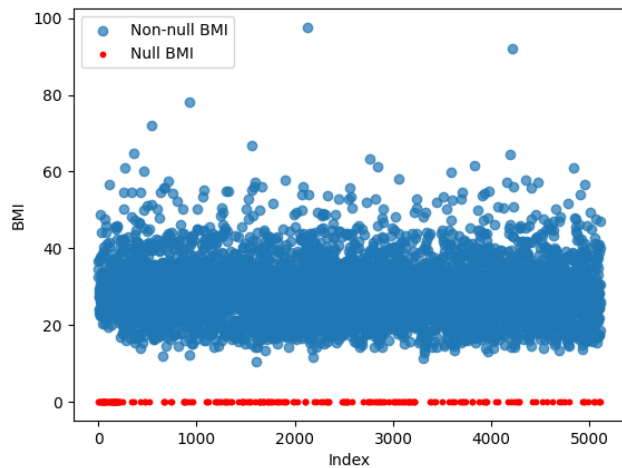
B. DATA IMPUTATION

While investigating the dataset further, and paying particular attention to the features, we identified that the ‘bmi’ field is missing data in roughly 4% of cases. For these missing values, we used a Random Forest Regressor technique to do imputations. This was because the null values were not huddled in a particular field as tend to happen sometimes, but were randomly distributed in the entire dataset. The first plot shows the ‘bmi’ values in the test set, with nodes in red denoting missing values and nodes in blue denoting actual values. The second graph demonstrates the new “bmi” values integrated by imputation, signified in green. These plots will give us an overview of how data has been imputed and its effect on ‘bmi’ column in general data.

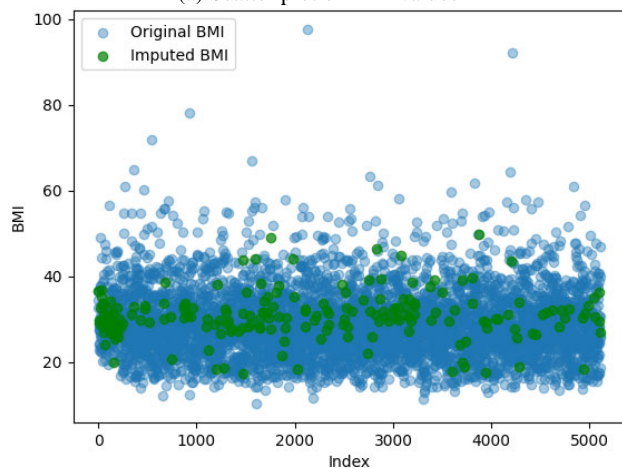
Furthermore, we also performed a detailed statistical analysis on the “bmi” column, before and after the imputation process was performed on it, to meet the fourth requirement. This entails calculating the average (mean), mid-point (median) and most frequent value (mode) of the “bmi” values. Our analysis provided evidence that these statistical measures were not affected by the process of imputation, that means our imputation method served effectively and did not influence the dataset by introducing biases. The exact figures of mean median as well as mode before as well as after the run of the imputation data is depicted in the plot as well. This is a very demanding statistical work, which helps to maintain the quality of data pre-processing and the accuracy of the further predictive models.

C. CORRELATION ANALYSIS

To better understand the relationships among features in our dataset, a correlation analysis was conducted using a heatmap representation, as shown in Figure 4. This analysis helps



(a) Scatter plot of BMI values



(b) Scatter plot of original and imputed BMI values

FIGURE 2. (a) Plot of BMI values in original dataset (b) Plot of BMI values after imputation.

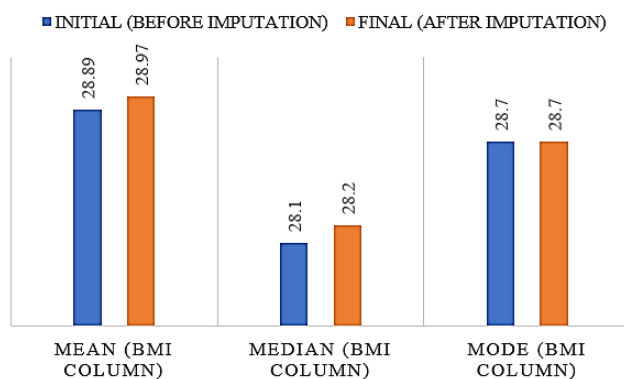


FIGURE 3. Mean, median and mode values before and after the imputation.

in identifying potentially influential factors contributing to stroke risk.

The correlation results indicate that no single feature exhibits a particularly strong correlation with stroke

occurrence, suggesting that stroke risk is influenced by a combination of multiple factors rather than a single dominant predictor. While age, hypertension, and average glucose level show moderate correlations with stroke, other features such as BMI and smoking status display weaker associations. This aligns with existing medical literature, where stroke risk is understood to be multifactorial, rather than dependent on one or two dominant predictors.

Since none of the features exhibit excessive collinearity, we retained all variables in the dataset to preserve the integrity of the model's learning process. This ensures that the predictive model leverages a diverse set of features rather than relying on any single variable.

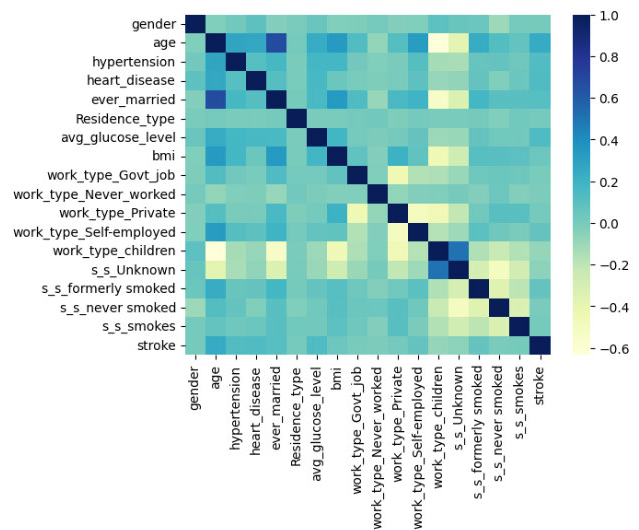


FIGURE 4. Correlation heatmap in different data columns.

D. ADDRESSING DATA IMBALANCE

One of the critical challenges in predictive modeling for stroke detection is the severe class imbalance present in the dataset. Since stroke cases account for only about 5% of the total data, machine learning models tend to be biased toward the majority class (non-stroke), leading to suboptimal performance in detecting actual stroke-positive instances. This section outlines the strategies employed to mitigate this issue and improve model reliability.

1) DEALING WITH DATA IMBALANCE

Data imbalance significantly affects classification models, as many algorithms prioritize overall accuracy rather than correctly classifying minority instances. In the case of stroke risk prediction, failing to correctly identify even a small number of stroke cases can have severe clinical consequences. Traditional models trained on imbalanced data may exhibit high accuracy but suffer from poor recall for the minority class, which is unacceptable in healthcare applications.

To counteract this issue, we explored various oversampling techniques aimed at increasing the representation of stroke cases in the training set while preserving generalizability.

2) IMPACT ON MODEL PERFORMANCE

When trained on highly imbalanced data, machine learning models tend to optimize for majority-class accuracy, leading to poor sensitivity (recall) for stroke cases. This results in a model that performs well in general but fails to identify individuals at risk.

To ensure a fair evaluation, we prioritized performance metrics beyond accuracy:

- **Precision:** Measures how many predicted stroke cases were actual strokes.
- **Recall (Sensitivity):** Determines how well the model identifies stroke cases.
- **F1-Score:** A harmonic mean of precision and recall, balancing both factors.
- **AUC-ROC (Area Under the Curve - Receiver Operating Characteristic):** Evaluates the trade-off between sensitivity and specificity, providing an overall measure of classification performance.

By focusing on these metrics, we ensured that our model was clinically relevant and not just numerically optimized.

3) OVERSAMPLING STRATEGIES

To mitigate class imbalance, we experimented with three primary oversampling techniques, applied only to the training set to prevent data leakage:

- 1) **Random Oversampling (ROS)**
 - This method duplicates existing minority-class samples until a desired class ratio is achieved.
 - While simple, it increases the risk of overfitting by repeatedly copying identical stroke cases.
- 2) **SMOTE (Synthetic Minority Over-sampling Technique)**
 - Instead of duplicating, SMOTE generates synthetic stroke cases by interpolating between real minority-class instances.
 - This approach creates more diverse training samples, helping the model learn meaningful stroke patterns while reducing overfitting risks.
- 3) **ADASYN (Adaptive Synthetic Sampling)**
 - Similar to SMOTE but places more emphasis on hard-to-learn stroke cases.
 - ADASYN adaptively generates synthetic samples in regions where the model struggles, potentially improving minority-class detection.

Each oversampling technique was combined with different feature scaling methods (Standard Scaling, Min-Max Scaling, and Robust Scaling) to evaluate their impact on model performance.

4) KEY OBSERVATIONS

- **Improved Minority-Class Recall:** Both SMOTE and ADASYN significantly enhanced stroke case detection, increasing recall for the positive (stroke) class.

- **Balancing Precision & Recall:** While oversampling helped capture more stroke cases, excessive oversampling increased false positives. We carefully balanced precision and recall to select the most effective technique.
- **Best Performing Model Configuration:** After extensive experimentation, we found that SMOTE combined with Standard Scaling in Logistic Regression provided the best balance of precision, recall, and stroke-class F1-score.
- **Clinical Relevance:** The optimized model ensures that high-risk individuals are not overlooked, improving the potential for early stroke prevention.

This systematic approach to handling data imbalance was crucial for achieving an equitable and reliable predictive model, ensuring that stroke risk assessments remain accurate and actionable in real-world healthcare applications.

E. DATA SEGREGATION

After the feature extraction from the data, it was necessary to split the attributes between features X and target variable y in our code. We also split the data into training and testing sets, where 80% of the data were used in train the model and 20% in test the model. This was done through the help of split function from the scikit-learn Python library, with a defined ratio of 0.2 through the “test_size” function, a random seed of 42 and “stratify” set to y .

Looking at the specific features of the introduced parameters for our context, the “stratify” parameter can be highlighted as the most significant. First, when we set the option ‘stratify’, we ensure that it keeps the distribution of labels i.e. In the context of this step the generalization between the values of the target variable remains constant between the calibration and validation phase. This is especially important in cases where the dataset is unbalanced or skewed, as it might be in this case where certain labels can be greatly outnumbered. As mentioned earlier, we had a total of 5109 observations and before applying the attribute icon, only 249 instances contained a value of “1” in the stroke attribute which constitutes the majority class. Thus, the fairly simple action of preserving labels’ distribution is highly useful for the mitigation of overfitting to the large class and improving models’ generalizability. It allows strengthening the reliability of the results of the further analyses obtained in the course of the study.

From figure 5, it can be observed that the ratio of the ‘1’ stroke values and the ‘0’ stroke values among the data before splitting and the data used for the training is coherent.

F. MODELS AND APPROACH

In our comparative part, we took a wide outlook on such models that include some basic models in the areas of Machine Learning as well as some important Deep Learning networks. The motivations for using these specific models are founded on its applicability and effectiveness within several operating functions within the ML. For the traditional ML

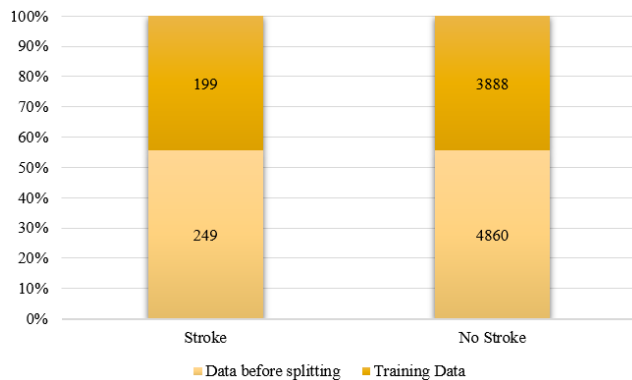


FIGURE 5. Data samples for each class before and after splitting.

models, it may be chosen because of its exposable and stable for the selected models, while for Deep learning models, there exist many reasons for selecting such models such as for the ability to solve non-linear relationships. In addition, parametric tuning was done with the aim of enhancing the optimum possible performance for the Machine Learning models.

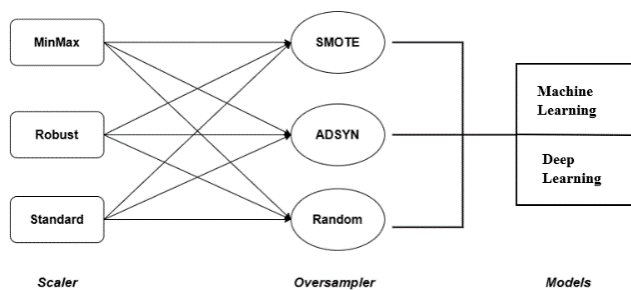


FIGURE 6. Flowchart showing the combination of scalers, oversamplers and models used.

Moreover, this last characteristic presents another useful application whether or not it is possible to combine the models mentioned above in machine learning, since it depicts a vast area in this domain. It also avoids the bias that may be caused by choosing certain model or learning algorithm, thus increasing the credibility and universalities of the report.

As pointed in Table 2, for all the Machine Learning models, we assigned the ‘scoring’ parameter as ‘accuracy’ or ‘precision’ depending on the applicability. For the Deep Learning models, the training process was done for 25 epochs each, and the batch size used was 32.

Since the dataset was highly deviated towards the negative target class (only 249 values for Positive i.e. Stroke class out of the 5109 total values considered), Each model was trained separately on three different sub-datasets, each corresponding to a different oversampling technique: The following synthetic methods will be used in the program: SMOTE, ADASYN, and Random Oversampling. Following oversampling, each sub-dataset was scaled using three different scalers: Previously, three types of data pre-

processing techniques were discussed; these are MinMax, Robust and Standard Scaler. This was done to determine which scaler, oversampler, and model combination would perform best on the assessed test data. This exercise in comparative analysis in truth covered a range of options and variations that was diverse and highly specific in every aspect of the modelling approach we aimed to adopt. The choice of SMOTE oversampling and Standard Scaler was determined after extensive experimentation with multiple oversampling techniques (SMOTE, ADASYN, and Random Oversampling) and scalers (Standard Scaler, MinMax Scaler, and Robust Scaler). SMOTE was preferred because it effectively generates synthetic minority class samples without introducing excessive noise, which is crucial for imbalanced datasets in healthcare applications. Additionally, Standard Scaler was selected due to its ability to preserve the distribution of features, ensuring that models generalize better. Considering the non-temporal, tabular nature of the dataset, we focused on non-recurrent machine learning models better suited to this classification task. Models such as Logistic Regression, Random Forest, and Support Vector Machines are selected for their established performance with static datasets, however still the methodology includes Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks for just benchmarking them on this data with our parameters to facilitate future research on various other datasets. The tool’s verification process emphasizes model accuracy and reliability by employing cross-validation and tuning for F1-scores in minority classes, specifically designed for healthcare applications. Unlike general-purpose prediction models, this tool’s validation approach ensures predictive consistency and adapts well to typical healthcare data irregularities, supporting its usability for accurate health risk assessment. In terms of scalability, while the current implementation of our web interface uses Streamlit, large-scale deployment would require backend optimizations such as API-based architectures, caching mechanisms, and cloud deployment. Future work could explore integrating a more scalable infrastructure to support higher user volumes.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

The original data for our analysis was our chosen data set, that is, Stroke Prediction data set from Kaggle. The target variable for output in this case is the patient’s likelihood of having a stroke and the input features include id, gender, age, various anomalies, marital status, work type, residence type, average glucose level, BMI and smoking status. Each row contains patient information concerning the problem. The raw dataset contained 12 columns of attributes and of 5110 number of observations in all. All these factors were found to have influence on stroke occurrence [23], [24], [25], [26], [27], [28], [29], [30], [31], [32]. This data set was pre-processed and then taken through model training where it was divided into a training data set and a testing data set. The constitution part of the raw data is also illustrated in figure 1.

TABLE 2. A table depicting different types of models and the parameters on their initialization used in this study.

Model Name	Parameters on initialization
Logistic Regression	random_state = 42, class_weight='balanced'
Logistic Regression (Gridsearched)	random_state = 42, class_weight='balanced' param_grid_lr = {'penalty': ['l1', 'l2'], 'C': [0.001, 0.01, 0.1, 1, 10], 'solver': ['liblinear', 'saga'], 'max_iter': [100, 200, 300]} cv = 5
Decision Trees (Normal)	random_state = 42, class_weight='balanced'
Decision Trees (Gridsearched)	random_state = 42, class_weight='balanced' param_grid_dt = {'criterion': ['gini', 'entropy'], 'max_depth': [None, 10, 20, 30, 40], 'min_samples_split': [2, 5, 10], 'min_samples_leaf': [1, 2, 4]} cv = 5
Random Forest (Normal)	random_state = 42, class_weight='balanced'
Random Forest (Gridsearched)	random_state = 42, class_weight='balanced' param_grid_rf = {'n_estimators': [50, 100, 200], 'max_depth': [None, 10, 20], 'min_samples_split': [2, 5, 10], 'min_samples_leaf': [1, 2]} cv = 5
SVM (Normal)	random_state = 42, class_weight='balanced'
SVM (Support Vector Machines) (Gridsearched)	random_state = 42, class_weight='balanced' param_grid_svm = {'C': [0.1, 1, 10], 'kernel': ['linear', 'rbf', 'poly'], 'gamma': ['scale', 'auto']} cv = 5
Gaussian Naïve Bayes	Nil
K-Nearest Neighbours (Normal)	Nil
K-Nearest Neighbours (Gridsearched)	param_grid_knn = {'n_neighbors': [3, 5, 7, 9], 'weights': ['uniform', 'distance'], 'p': [1, 2]} cv = 5
Artificial Neural Network	ann_model = Sequential([Dense(128, activation='relu', input_dim=X_train_scaled.shape[1]), Dropout(0.5), Dense(64, activation='relu'), Dense(1, activation='sigmoid')]) Compiled with: optimizer='adam', loss='binary_crossentropy', metrics=['accuracy']
1D – Convolutional Neural Network	Reshaping the input data. cnn_model = Sequential([Conv1D(32, 3, activation='relu', input_shape=X_train_cnn.shape[1:]), MaxPooling1D(2), Flatten(), Dense(128, activation='relu'), Dropout(0.5), Dense(1, activation='sigmoid')]) Compiled with: optimizer='adam', loss='binary_crossentropy', metrics=['accuracy']
Bi-directional Recurrent Neural Network	birnn_model = Sequential([Bidirectional(LSTM(64, activation='relu', return_sequences=True), input_shape=X_train_rnn.shape[1:]), Dropout(0.5), Bidirectional(LSTM(32, activation='relu')), Dropout(0.5), Dense(1, activation='sigmoid')]) Compiled with: optimizer='adam', loss='binary_crossentropy', metrics=['accuracy']
Long Short Term Memory (LSTM)	lstm_model = Sequential([LSTM(64, activation='relu', input_shape=X_train_rnn.shape[1:], return_sequences=True), Dropout(0.5), LSTM(32, activation='relu'), Dropout(0.5), Dense(1, activation='sigmoid')]) Compiled with: optimizer='adam', loss='binary_crossentropy', metrics=['accuracy']
Gated Recurrent Unit (GRU)	gru_model = Sequential([GRU(64, activation='relu', input_shape=X_train_rnn.shape[1:], return_sequences=True), Dropout(0.5), GRU(32, activation='relu'), Dropout(0.5), Dense(1, activation='sigmoid')]) Compiled with: optimizer='adam', loss='binary_crossentropy', metrics=['accuracy']

A. PERFORMANCE METRICS

In this section, we discuss the performance metrics used to evaluate the various machine learning and deep learning models in our study. The goal was to identify the best model combination for predicting an individual's vulnerability to brain stroke. We focused on three primary metrics: Accuracy, $F_{1,0}$ score for the no-stroke class, and $F_{1,1}$ score for the stroke class. These metrics provide a comprehensive view

of model performance, particularly focusing on the minority class (stroke), which is crucial for medical predictions.

1) ACCURACY

Accuracy measures the proportion of correctly classified instances out of the total instances:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

where:

- *TP* (True Positives): Correctly predicted positive cases.
- *TN* (True Negatives): Correctly predicted negative cases.
- *FP* (False Positives): Incorrectly predicted positive cases.
- *FN* (False Negatives): Incorrectly predicted negative cases.

Accuracy provides an overall performance measure but can be misleading in imbalanced datasets, which is why additional metrics are necessary.

2) $F_{1,0}$ SCORE FOR NO-STROKE CLASS

The $F_{1,0}$ score is the harmonic mean of precision and recall for the no-stroke class:

$$F_{1,0} = 2 \cdot \frac{\text{Precision}_0 \cdot \text{Recall}_0}{\text{Precision}_0 + \text{Recall}_0}$$

where:

$$\text{Precision}_0 = \frac{TN}{TN + FN}, \quad \text{Recall}_0 = \frac{TN}{TN + FP}$$

$F_{1,0}$ balances the trade-off between precision and recall, ensuring that both false positives and false negatives are considered for the no-stroke class.

3) $F_{1,1}$ SCORE FOR STROKE CLASS

The $F_{1,1}$ score is the harmonic mean of precision and recall for the stroke class:

$$F_{1,1} = 2 \cdot \frac{\text{Precision}_1 \cdot \text{Recall}_1}{\text{Precision}_1 + \text{Recall}_1}$$

where:

$$\text{Precision}_1 = \frac{TP}{TP + FP}, \quad \text{Recall}_1 = \frac{TP}{TP + FN}$$

$F_{1,1}$ is particularly important in our study because it evaluates the model's performance in correctly identifying stroke cases, which is critical for preventive healthcare.

By optimizing these measures, we guarantee the accomplishment of two objectives simultaneously: the overall accuracy of the selected model and its ability to correctly identify stroke cases, even within a small intersection of the dataset.

B. RESULTS AND DISCUSSION

Our work included detailed pairwise comparison of 144 cases, including 99 ML instances (11 distinct ML models with 3 oversampling techniques and 3 scalers) and 45 DL instances with five various DL models. There were many types of models trained and tested: logistic regression, random forest classifier, and support vector machines classifier, and many more, each of which was applied to the three oversampling techniques that included SMOTE, ADASYN, and random oversampling and combined with three data scalers including MinMax Scaler, Robust Scaler, and Standard Scaler.

Therefore, based on the analysis carried out, there were considerable performances' differences between the various models, the oversampling techniques and the scalers. The dataset's non-sequential nature made non-recurrent models more suitable. Therefore, our final model selection excluded LSTM and GRU in favor of conventional models optimized for tabular data.

To provide a comparative perspective on stroke risk prediction, we include a performance comparison table summarizing key evaluation metrics of our model alongside prior research. This comparison highlights differences in dataset size, handling of class imbalance, feature preprocessing, and model performance metrics (Accuracy, Precision, Recall and F1-score).

Several studies have explored stroke prediction using different machine learning and deep learning techniques. Table 6 presents a comparative analysis of our proposed model against prior works in terms of dataset size, preprocessing techniques, oversampling methods, and model performance. To shed more light on evaluating the effectiveness of our LR + Standard Scaler + SMOTE model, we compared its performance against several state-of-the-art stroke prediction systems and with the addition of SOTA method called "Intelligent System (IS)". Table 6 presents a detailed comparison across key metrics, highlighting the relative strengths and trade-offs of each approach.

The IS achieves superior performance, boasting 99.24% accuracy and a 99.00% F1 score, surpassing our LR + Standard Scaler + SMOTE model by approximately 6% in accuracy and 4 points in F1. This advantage stems from two innovations: compact, information-preserving feature compression via autoencoders, and a balanced training regime using latent-space SMOTE to enhance minority recall. However, IS demands significant computational resources, including GPU pretraining and LDA solver optimization. In contrast, our lightweight LR pipeline offers fast, resource-efficient inference, making it highly suitable for real-time browser deployment, while the IS can support deeper backend validation.

Our results demonstrate that handling class imbalance with SMOTE and using Standard Scaler significantly improves stroke prediction performance, particularly in terms of recall and F1-score, which are crucial for healthcare applications. Unlike prior studies that either neglected data imbalance or relied solely on accuracy, our approach prioritizes recall, ensuring a clinically relevant balance between sensitivity and specificity.

This evaluation demonstrated that it is hard to predict a stroke and that outcomes varied depending on the chosen model, over-sample method, and scaler applied option, therefore, the choice of the correct strategy is crucial for high performance. As can be seen, one significant factor that emerged from this study is the continuing relevance of the comparative assessments of a wide variety of methods when it comes to developing predictive models. Using the feature of grid search, four principles of models, two types of methods

TABLE 3. Comparison of accuracy for machine learning model.

Model	Scaler	Oversampler	Accuracy	F1_0	F1_1
LR_GridSearched	Standard	SMOTE	0.932485	0.964341	0.366972
LR_Normal	MinMax	ADASYN	0.936399	0.966547	0.356436
1DCNN	Robust	SMOTE	0.898239	0.945378	0.257143
KNN_GridSearched	Robust	SMOTE	0.917808	0.956522	0.25
KNN_Normal	Robust	ADASYN	0.910959	0.952727	0.235294
SVM_NORMAL	MinMax	Random	0.751468	0.851809	0.230303
GRU	Robust	SMOTE	0.906067	0.95	0.225806
ANN	Robust	Random	0.793542	0.880993	0.221402
DT_Normal	Robust	SMOTE	0.913894	0.954451	0.214286
DT_GridSearched	Standard	Random	0.912916	0.95391	0.212389
LSTM	MinMax	Random	0.706458	0.819712	0.210526
SVM_GridSearched	MinMax	Random	0.793542	0.881527	0.197719
BiRNN	Robust	Random	0.661448	0.785626	0.195349
GaussianNB	Robust	SMOTE	0.419765	0.56493	0.129222
RF_GridSearched	Robust	Random	0.946184	0.972264	0.098361
RF_Normal	Standard	SMOTE	0.940313	0.969145	0.089552

TABLE 4. Comparison of accuracy for deep learning model.

Model	Scaler	Oversampler	Accuracy	F1_0	F1_1
1DCNN	Robust	SMOTE	0.898239	0.945378	0.257143
GRU	Robust	SMOTE	0.906067	0.95	0.225806
ANN	Robust	Random	0.793542	0.880993	0.221402
LSTM	MinMax	Random	0.706458	0.819712	0.210526
BiRNN	Robust	Random	0.661448	0.785626	0.195349

TABLE 5. Comparison of accuracy for all the best performing models from our paper.

Model	Scaler	Oversampler	Accuracy	F1_0	F1_1
LR_GridSearched	Standard	SMOTE	0.932485	0.964341	0.366972
LR_Normal	MinMax	ADASYN	0.936399	0.966547	0.356436
1DCNN	Robust	SMOTE	0.898239	0.945378	0.257143
KNN_GridSearched	Robust	SMOTE	0.917808	0.956522	0.25
KNN_Normal	Robust	ADASYN	0.910959	0.952727	0.235294
SVM_NORMAL	MinMax	Random	0.751468	0.851809	0.230303
GRU	Robust	SMOTE	0.906067	0.95	0.225806
ANN	Robust	Random	0.793542	0.880993	0.221402
DT_Normal	Robust	SMOTE	0.913894	0.954451	0.214286
DT_GridSearched	Standard	Random	0.912916	0.95391	0.212389
LSTM	MinMax	Random	0.706458	0.819712	0.210526
SVM_GridSearched	MinMax	Random	0.793542	0.881527	0.197719
BiRNN	Robust	Random	0.661448	0.785626	0.195349
GaussianNB	Robust	SMOTE	0.419765	0.56493	0.129222
RF_GridSearched	Robust	Random	0.946184	0.972264	0.098361
RF_Normal	Standard	SMOTE	0.940313	0.969145	0.089552

for oversampling, and five scalers, it is possible to determine which detected result of models in the test set was found to be the best. For this study, the model is described as the LR_Gridsearched SMOTE-Standard model, the one with the highest F1 score in the minority class. The table summarizes the performance across multiple metrics, emphasizing the Logistic Regression model with SMOTE and Standard Scaler for its consistently high scores in accuracy and F1-score. This comprehensive metric evaluation highlights the model's robustness in distinguishing stroke risk, particularly

in imbalanced datasets. For the reference of the readers, additional metric results such as Precision and recall of the models can be accessed on the following GitHub repository <https://github.com/bh-divyansh/Brain-Stroke-Results>.

This is normal because our dataset was imbalanced; more instances of non-stroke (0) and less instances of strokes (1) were collected, therefore the F1 Score for the positive class, F1_1 was used to determine which amongst the two models is most suitable to develop the web application. The best cross-validation, which achieved the highest F1_1 [for

TABLE 6. Comparison of stroke prediction models on test data using Accuracy, Precision, Recall, and F1 Score. * Intelligent System (IS): Saleem, M.A., Javeed, A., Akarathanawat, W. et al. An intelligent learning system based on electronic health records for unbiased stroke prediction. Sci Rep 14, 23052 (2024). <https://doi.org/10.1038/s41598-024-73570-x>.

Model	Accuracy (%)	Precision (%)	Recall / Sensitivity (%)	F1-Score (%)
[33] Traditional SVM (IST)	75.08	98.91	84.00	90.84
[34] Decision Tree + PCA + NN	94.11	93.27	95.06	94.16
[35] Random Forest (NIHSS)	97.00	95.00	99.00	97.00
Intelligent System (IS)*	99.24	99.50	98.51	99.00
Our LR + Standard Scaler + SMOTE	93.25	96.88	93.25	94.99*

the minority class] of 0. LR_Model: 36697 is the name of the Grid-searched Logistic Regression (LR_Gridsearched) model that was used in combination with standard scaler data normalization technique and with SMOTE oversampling method. As such, its incorporation into the web application used in this approach was adopted because it can be used to prescreen any person wishing to determine their stroke risk status.

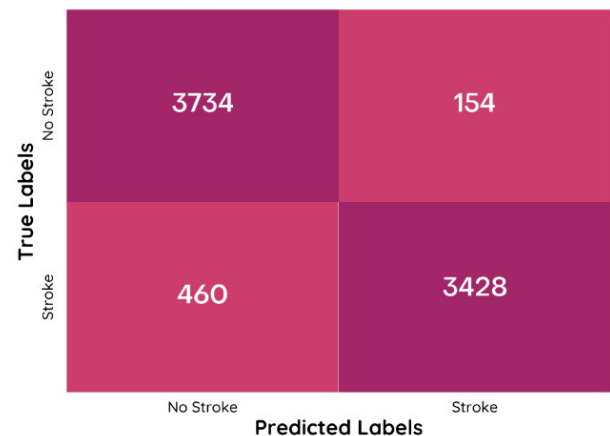
The web interface was developed using the Python library, Streamlit, which enabled us to design a user-friendly interface. We initiated the process by generating a template from the 'streamlit-hello' app, as per the official Streamlit documentation. This template was then customized on GitHub Codespaces to create our web interface. The web interface can be accessed via <https://strokedetection.streamlit.app/>.

The web interface was developed using the Python library, Streamlit, which enabled us to design a user-friendly interface. We initiated the process by generating a template from the 'streamlit-hello' app, as per the official Streamlit documentation. This template was then customized on GitHub Codespaces to create our web interface. The web interface can be accessed via <https://strokedetection.streamlit.app/>.

The interface for performing our search is user-friendly and simple; indeed, it resembles filling out an online questionnaire. Subsequently, all the data regarding the user such as age, BMI, work type and many others can easily be entered through Drop Down Boxes in the Graphical User Interface. To maintain user privacy all the code operation is done directly in real time without a back end Database, hence no database marks are made, no data is collected or stored. The web-based tool ensures secure real-time processing by handling all user inputs locally, meaning no data is stored or transmitted. This design prioritizes user privacy, eliminating concerns about data leaks.

However, in future implementations, additional security measures such as end-to-end encryption and secure authentication could be introduced to further enhance privacy. Additionally, integrating federated learning techniques would allow continuous model improvement while maintaining data confidentiality. User inputs need to be scaled in the same way as the inputs that were used during the model training process, and the scaled inputs need to be passed through the selected model. This means that the output of the model is presented such that the user can understand its presentations to make the necessary actions. To make the site accessible, we use Streamlit Share as our hosting and deployment platform, also

Training Data Confusion Matrix



Testing Data Confusion Matrix

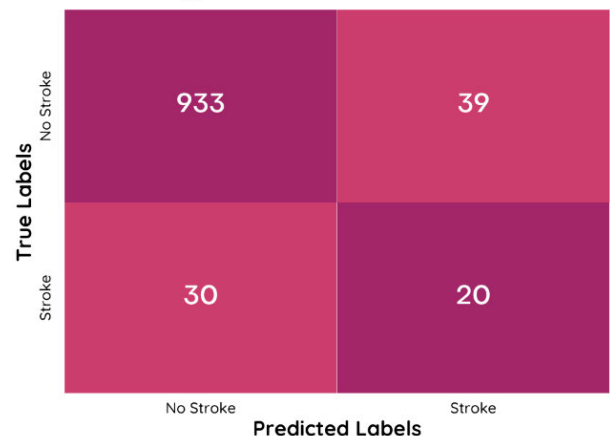


FIGURE 7. Confusion matrix for final model on chosen (a) Scaler (b) Oversampler used in our web tool.

known as the Streamlit Community Cloud. If there are any cookie data generated through the interface, the platform ensures their management, and the Data Privacy Policy will apply. As per the authors, in the interface, they do not keep any data of the users in any way and shape. A diagram illustrating the user-end process is provided in Fig 9.

V. ETHICAL CONSIDERATIONS AND POTENTIAL BIASES

Like any healthcare-focused machine learning application, the proposed stroke risk assessment tool must navigate several ethical challenges. First, there is the risk of overreliance on algorithmic output patients or clinicians might treat a

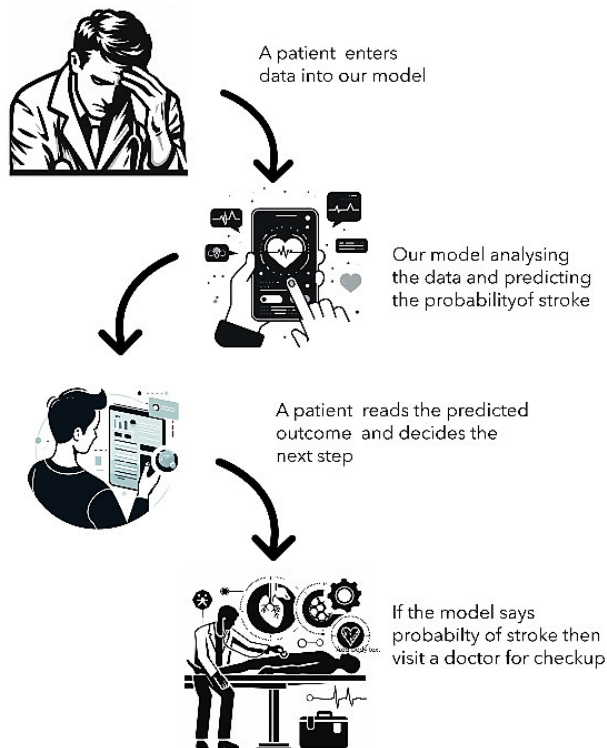


FIGURE 8. A diagram illustrating the user-end process.

“low-risk” result as definitive, delaying important medical follow-ups. To mitigate this, the interface clearly states that its predictions are merely advisory and should be confirmed by professional consultation. Second, biases can emerge if the training data underrepresents certain demographic groups (e.g., younger individuals, certain ethnicities, or those in rural areas). This could inadvertently lead to skewed predictions in populations not well-represented in the original dataset. SMOTE and similar oversampling techniques address class imbalance but do not automatically correct for demographic or other sampling biases. Third, data privacy and informed consent matter greatly in healthcare: while this interface does not store user inputs, organizations adopting the tool in clinical settings must ensure compliance with relevant regulations (e.g., HIPAA, GDPR) if data collection or logging is introduced. Finally, interpretability becomes a core ethical requirement users should understand at least the basic rationale behind a model’s predictions, or risk placing undue trust in a “black box.” By using a more transparent model like logistic regression and emphasizing disclaimers, the tool aims to reduce harm from misinterpretation or overconfidence in algorithmic judgments.

VI. PRACTICAL IMPLICATIONS FOR REAL-WORLD DEVELOPMENT

The preference for simpler, interpretable models, specifically logistic regression offers direct benefits for public health applications. Unlike black-box deep learning architectures,

logistic regression yields relatively transparent insights into how each feature (e.g., BMI, blood pressure, smoking status) influences stroke risk. In a real-world context, this interpretability can foster greater trust among clinicians and end-users, who can more readily grasp why a particular risk level is determined. Further, the web-based interface was deliberately designed with non-technical users in mind: drop-down menus, intuitive instructions, and real-time feedback eliminate the need for advanced training or specialized software. Patients, caregivers, and healthcare workers can simply enter relevant health metrics like age, gender, and glucose levels and receive an immediate, comprehensible risk assessment. This accessibility is vital in community or remote settings, where medical resources or technical expertise may be limited. By not requiring any software installation or data-storage infrastructure, the tool also alleviates concerns around deployment overhead and privacy. Ultimately, adopting an easy-to-understand model and a straightforward web interface broadens the tool’s potential user base enabling earlier intervention for at-risk individuals who might otherwise lack a clear pathway to risk evaluation.

VII. PERFORMANCE COMPARISON WITH STATE OF THE ART (SOTA)

When we trained a Logistic-Regression + StandardScaler + SMOTE model on the same Kaggle dataset and kept oversampled records strictly inside the training folds performance reached 93.25 % accuracy and 94.99 % F1. That is < 6 percentage points behind the IS but with:

- 1) One-line interpretability (feature odds-ratios).
- 2) CPU-level inference (no GPU or autoencoder pre-training).
- 3) Lower risk of metric inflation, because LR lacks the capacity to overfit synthetic artefacts as readily as a deep latent model.

Hence, although the IS state-of-the-art, its incremental gain may not justify the added complexity and potential over-optimism unless external validation on genuinely independent cohorts confirms the same advantage. Despite the clear quantitative gain, the IS still requires GPU assisted pre training and a grid search over LDA solvers, making our lightweight LR stack attractive for resource constrained web deployment. A pragmatic workflow might therefore run LR SMOTE in the browser for rapid triage and refer borderline cases (decision threshold 0.40–0.60) to the heavier IS back end hosted on a server. To overcome the issues of achieving inflated scores, we argue that an ablation study for in-depth analysis must be carried out. While the autoencoder + LDA + SMOTE pipeline posts headline figures of 99 %-plus for accuracy, precision, recall, and F1, those metrics may be inflated because the evaluation protocol does not fully guard against two sources of optimism:

- 1) **Synthetic-sample bias.** SMOTE injects artificial minority-class observations that are linearly interpolated from existing stroke cases. If the same interpolation manifold is present in both training and test

folds (e.g., under random shuffling without grouped or temporal splits), the classifier will “see” patterns during testing that are almost identical to what it learned, artificially boosting recall and F1.

- 2) **Class-imbalance masking.** Accuracy > 99 % is achievable even with a naïve majority-class predictor when stroke prevalence is < 5 %. Precision can also be overstated if true-negative counts dominate the contingency table. A more stringent assessment would report Matthew’s correlation coefficient, balanced accuracy, PR-AUC, and calibration plots, and would evaluate on a hold-out set untouched by any oversampling.

VIII. LIMITATIONS

While the proposed model achieves high accuracy, several limitations must be acknowledged. The dataset consists of only 5,110 samples, with a severe class imbalance—only 249 stroke-positive cases. Although SMOTE (Synthetic Minority Over-sampling Technique) was employed to generate synthetic samples and mitigate this imbalance, the dataset’s limited size may still impact the model’s generalizability. Larger, more diverse datasets from multiple sources would be necessary to further validate the model’s robustness.

Additionally, the model has not undergone clinical validation, meaning its predictions should be interpreted with caution. While it can serve as a preliminary assessment tool, it is not a substitute for professional medical diagnosis. Future research should focus on extensive clinical trials to establish the model’s reliability in real-world healthcare settings, ensuring its predictions align with medical standards.

Extensive experimental comparisons revealed that SMOTE and Standard Scaler consistently outperformed other pre-processing techniques when evaluated on critical healthcare metrics such as recall, F1-score, and AUC-ROC. SMOTE generates synthetic data points by interpolating between real minority-class instances, avoiding the pitfalls of random oversampling (which risks overfitting by duplicating rare cases) and ADASYN (which can sometimes generate overly specialized samples in borderline regions). In a high-stakes domain like stroke prediction, where missing positive cases can have severe consequences, SMOTE provides a stable mechanism to enhance minority-class detection without artificially inflating false positives.

Similarly, Standard Scaler (Z-score normalization) was chosen over MinMax Scaling and Robust Scaling due to its ability to preserve the underlying distribution of features. While MinMax scaling maps values into a fixed range [0,1], potentially exaggerating outliers, and Robust Scaling trims outliers based on interquartile ranges, Standard Scaler ensures each feature contributes comparably to the model without distorting meaningful variations. In a medical dataset where key biomarkers (e.g., BMI, glucose levels, blood pressure) often follow a near-Gaussian distribution, this normalization method maintains critical patterns without diminishing clinically relevant outlier information. The

synergy between SMOTE and Standard Scaler ultimately leads to more reliable, clinically relevant predictions.

Finally, the scalability of the web-based interface poses another challenge. The current implementation is optimized for individual users, but real-world deployment in hospitals or large-scale clinical environments would require backend optimization and seamless integration with electronic health records (EHR). Future work could explore cloud-based solutions to handle high-volume traffic, ensuring efficient data processing and real-time stroke risk assessment.

IX. CONCLUSION

In conclusion, this research has successfully developed and tested a fully operational web-based interface designed to assess an individual’s risk of brain stroke, a leading global health concern. The interface utilizes a machine learning model trained on a Kaggle dataset, where an extensive evaluation was conducted using 11 machine learning algorithms and 5 deep learning algorithms, each tested with different scalers and oversampling techniques. After comparative analysis, the most suitable model architecture—Logistic Regression with SMOTE oversampling and Standard Scaler—was selected due to its balanced trade-off between interpretability, accuracy, and computational efficiency.

While this study primarily focuses on model performance and accessibility, future work should explore the integration of real-time health monitoring data, such as wearable sensor metrics (heart rate, blood pressure fluctuations) to further improve predictive capabilities. Additionally, ensemble and deep learning-based hybrid models could be investigated to enhance predictive power while maintaining explainability for clinical applications.

One of the critical challenges in stroke prediction remains the interpretability of machine learning models. While Logistic Regression provides inherent feature transparency, incorporating explainable AI (XAI) techniques such as SHAP (Shapley Additive Explanations) could improve trust and usability for healthcare professionals. Moreover, usability testing with a diverse group of non-technical users would provide valuable insights into optimizing the interface for practical, real-world deployment.

From an ethical standpoint, while SMOTE has been effective in mitigating class imbalance, oversampling techniques may introduce biases that could affect the generalizability of predictions. Future research should explore clinically validated datasets and conduct external validations across diverse populations to improve model fairness and reliability. Additionally, while our system does not store or transmit any user data, further enhancements in security, such as encryption and federated learning approaches, could be considered for large-scale adoption.

Finally, while this tool provides a user-friendly approach to stroke risk assessment, it is not intended as a replacement for clinical diagnosis but rather as a supplementary tool for early risk identification. Expanding this system to integrate with electronic health records (EHRs) and telemedicine

platforms could improve accessibility for both individuals and healthcare professionals. With continued research in health informatics and AI-driven predictive modeling, this work serves as a stepping stone for future advancements in stroke prevention and early intervention strategies.

REFERENCES

- [1] Prevent Brain Stroke—Who.int. Accessed: May 29, 2024. [Online]. Available: <https://www.who.int/southeastasia/news/detail/29-10-2016-prevent-brain-stroke>
- [2] Stroke—Ninds.nih.gov. Accessed: May 29, 2024. [Online]. Available: <https://www.ninds.nih.gov/health-information/disorders/stroke#toc-who-is-more-likely-to-have-a-stroke>
- [3] M. J. Zigmond, C. A. Wiley, and M.-F. Chesselet, *Neurobiology of Brain Disorders: Biological Basis of Neurological and Psychiatric Disorders*. New York, NY, USA: Academic, 2022.
- [4] B. Kim, N. Schweighofer, J. P. Haldar, R. M. Leahy, and C. J. Winstein, “Corticospinal tract microstructure predicts distal arm motor improvements in chronic stroke,” *J. Neurologic Phys. Therapy*, vol. 45, no. 4, pp. 273–281, 2021.
- [5] M. Lee, J. Ryu, and D.-H. Kim, “Automated epileptic seizure waveform detection method based on the feature of the mean slope of wavelet coefficient counts using a hidden Markov model and EEG signals,” *ETRI J.*, vol. 42, no. 2, pp. 217–229, Apr. 2020.
- [6] K. Moulaei, L. Afshari, R. Moulaei, B. Sabet, S. M. Mousavi, and M. R. Afrash, “Explainable artificial intelligence for stroke prediction through comparison of deep learning and machine learning models,” *Sci. Rep.*, vol. 14, no. 1, p. 31392, Dec. 2024.
- [7] T. I. Shaily, T. Islam, S. Jannat, S. A. Tanna, T. M. Alif, and R. R. Ema, “Detection of stroke disease using machine learning algorithms,” in *Proc. 10th Int. Conf. Comput., Commun. Netw. Technol. (ICCCNT)*, Jul. 2019, pp. 1–6.
- [8] C.-Y. Chang, M.-J. Cheng, and M. H. Ma, “Application of machine learning for facial stroke detection,” in *Proc. IEEE 23rd Int. Conf. Digit. Signal Process. (DSP)*, Nov. 2018, pp. 1–5.
- [9] S. Kashi, R. F. Polak, B. Lerner, L. Rokach, and S. Levy-Tzedek, “A machine-learning model for automatic detection of movement compensations in stroke patients,” *IEEE Trans. Emerg. Topics Comput.*, vol. 9, no. 3, pp. 1234–1247, Jul. 2021.
- [10] Z. G. Al-Mekhlafi, E. M. Senan, T. H. Rassem, B. A. Mohammed, N. M. Makbol, A. A. Alanazi, T. S. Almurayziq, and F. A. Ghaleb, “Deep learning and machine learning for early detection of stroke and haemorrhage,” *Comput., Mater. Continua*, vol. 72, no. 1, pp. 775–796, 2022.
- [11] T. Malini, M. Deepalakshmi, B. Dhivyaa, P. Karthikeswari, and N. Kavipriya, “Advanced stroke detection and alert system using machine learning,” in *Proc. 7th Int. Conf. Commun. Electron. Syst. (ICCES)*, Jun. 2022, pp. 1084–1089.
- [12] A. N. Tusher, Md. S. Sadik, and Md. T. Islam, “Early brain stroke prediction using machine learning,” in *Proc. 11th Int. Conf. Syst. Model. Advancement Res. Trends (SMART)*, Dec. 2022, pp. 1280–1284.
- [13] K. S. Raja, S. B. Chandra, K. Kausalya, R. M. Ciddarth, and V. G. Ranjith, “Prognosis of stroke using machine learning algorithms,” in *Proc. 7th Int. Conf. Comput. Methodologies Commun. (ICCMC)*, Feb. 2023, pp. 1–6.
- [14] S. P. Singh, L. Wang, S. Gupta, H. Goli, P. Padmanabhan, and B. Gulyás, “3D deep learning on medical images: A review,” *Sensors*, vol. 20, no. 18, p. 5097, Sep. 2020.
- [15] M. Chawla, S. Sharma, J. Sivaswamy, and L. T. Kishore, “A method for automatic detection and classification of stroke from brain CT images,” in *Proc. Annu. Int. Conf. IEEE Eng. Med. Biol. Soc.*, Sep. 2009, pp. 3581–3584.
- [16] M. S. Sirsat, E. Fermé, and J. Câmara, “Machine learning for brain stroke: A review,” *J. Stroke Cerebrovascular Diseases*, vol. 29, no. 10, Oct. 2020, Art. no. 105162.
- [17] A. M. Barrett, O. Boukrina, and S. Saleh, “Ventral attention and motor network connectivity is relevant to functional impairment in spatial neglect after right brain stroke,” *Brain Cognition*, vol. 129, pp. 16–24, Feb. 2019.
- [18] D. A. Brenner, R. M. Zweifler, C. R. Gomez, B. M. Kissela, D. Levine, G. Howard, B. Coull, and V. J. Howard, “Awareness, treatment, and control of vascular risk factors among stroke survivors,” *J. Stroke Cerebrovascular Diseases*, vol. 19, no. 4, pp. 311–320, Jul. 2010.
- [19] S. Bacchi, T. Zerner, L. Oakden-Rayner, T. Kleinig, S. Patel, and J. Jannes, “Deep learning in the prediction of ischaemic stroke thrombolysis functional outcomes,” *Academic Radiol.*, vol. 27, no. 2, pp. e19–e23, Feb. 2020.
- [20] M. Puttagunta and S. Ravi, “Medical image analysis based on deep learning approach,” *Multimedia Tools Appl.*, vol. 80, no. 16, pp. 24365–24398, Jul. 2021.
- [21] A. AlSobeh, “OSM: Leveraging model checking for observing dynamic I behaviors in aspect-oriented applications,” 2024, *arXiv:2403.01349*.
- [22] Stroke Prediction Dataset—Kaggle.com. Accessed: May 29, 2024. [Online]. Available: <https://www.kaggle.com/datasets/fedesoriano/stroke-prediction-dataset>
- [23] M. J. Haley and C. B. Lawrence, “Obesity and stroke: Can we translate from rodents to patients?” *J. Cerebral Blood Flow Metabolism*, vol. 36, no. 12, pp. 2007–2021, Dec. 2016.
- [24] J. Miller, H. Kinni, C. Lewandowski, R. M. Nowak, and P. D. Levy, “Management of hypertension in stroke,” *Ann. Emergency Med.*, vol. 64, no. 3, pp. 248–255, Apr. 2014.
- [25] M. J. Cipolla, D. S. Liebeskind, and S.-L. Chan, “The importance of comorbidities in ischemic stroke: Impact of hypertension on the cerebral circulation,” *J. Cerebral Blood Flow Metabolism*, vol. 38, no. 12, pp. 2129–2149, Dec. 2018.
- [26] G. Faraco and C. Iadecola, “Hypertension: A harbinger of stroke and dementia,” *Hypertension*, vol. 62, no. 5, pp. 810–817, Nov. 2013.
- [27] S. Zhang, W. Zuo, X.-F. Guo, W.-B. He, and N.-H. Chen, “Cerebral glucose transporter: The possible therapeutic target for ischemic stroke,” *Neurochemistry Int.*, vol. 70, pp. 22–29, May 2014.
- [28] H. Jørgensen, H. Nakayama, H. O. Raaschou, and T. S. Olsen, “Stroke in patients with diabetes. The Copenhagen stroke study,” *Stroke*, vol. 25, no. 10, pp. 1977–1984, Oct. 1994.
- [29] A. E. Sifat, S. Nozohouri, S. R. Archie, E. A. Chowdhury, and T. J. Abbruscato, “Brain energy metabolism in ischemic stroke: Effects of smoking and diabetes,” *Int. J. Mol. Sci.*, vol. 23, no. 15, p. 8512, Jul. 2022.
- [30] B. T. Hawkins, R. C. Brown, and T. P. Davis, “Smoking and ischemic stroke: A role for nicotine?” *Trends Pharmacological Sci.*, vol. 23, no. 2, pp. 78–82, Feb. 2002.
- [31] J. Wang, X. Ning, L. Yang, J. Tu, H. Gu, C. Zhan, W. Zhang, and T.-C. Su, “Sex differences in trends of incidence and mortality of first-ever stroke in rural Tianjin, China, from 1992 to 2012,” *Stroke*, vol. 45, no. 6, pp. 1626–1631, Jun. 2014.
- [32] R. L. Sacco, B. Boden-Albala, R. Gan, X. Chen, D. E. Kargman, S. Shea, M. C. Paik, and W. A. Hauser, “Stroke incidence among white, black, and hispanic residents of an urban community: The northern Manhattan stroke study,” *Amer. J. Epidemiology*, vol. 147, no. 3, pp. 259–268, Feb. 1998.
- [33] N. Patil and A. Sumarsono, “Stroke prediction using machine learning,” *J. Res. Eng. Comput. Sci.*, vol. 2, no. 1, pp. 61–72, 2024.
- [34] O. Shobayo, O. Zachariah, M. O. Odusami, and B. Ogunleye, “Prediction of stroke disease with demographic and behavioural data using random forest algorithm,” *Analytics*, vol. 2, no. 3, pp. 604–617, Aug. 2023.
- [35] R. Tehseen, U. Omer, R. Javaid, M. Mehr, M. Yousaf, and A. Zaheer, “Prediction of brain stroke using federated learning,” *Int. J. Innov. Sci. Technol.*, vol. 6, pp. 1995–2013, Dec. 2024.



India's evolving landscape.

DIVYANSH BHANDARI is currently pursuing the Bachelor of Technology degree in electronics and computer engineering with Vellore Institute of Technology, Chennai, India. His academic pursuits are strongly aligned with the dynamic fields of artificial intelligence and machine learning, areas in which he holds a profound interest. With existing published research in this domain, he is eager to contribute to and observe the transformative potential of AI technology within



ARNAV AGARWAL is currently pursuing the B.Tech. degree in electronics and computer engineering with Vellore Institute of Technology, Chennai. He has an interest in modern advancements in artificial intelligence and machine learning, especially in healthcare. This is his second article related to the topic, and he wishes to contribute to the betterment of public healthcare. His first paper was accepted at the SPCOM conference. He further wants to see the growing advancements in healthcare towards making a sustainable solution for the betterment of the country.



RAJARAM PRIYATHARSHINI received the Ph.D. degree, in October 2017, with a thesis titled “An Efficient Mining Framework for Coronary Disease Diagnosis from Bi-Modal Clinical Data” from Anna University, Chennai, India. She has more than 20 years of experience in teaching and research. She is currently a Lecturer and the Program Director of Cardiff School of Technologies, Cardiff Metropolitan University, U.K. She has more than 75 bachelor’s/master’s students who have completed their thesis dissertations under her supervision. She has published more than 40 papers in international journals and conferences, accumulating more than 200 citations, and has contributed to five book chapters published by Springer and Elsevier Academic Press. Her research interests include applying AI technologies in healthcare, medical image analysis, and data science. She received the Best Project Award from Tamil Nadu Science and Technology (TNSCST) for the project titled “Demand-Based Blue-Collared Job Scheduling using Meta-Heuristics Approaches” in 2017. She was recognized as the Outstanding Woman in the field of Computer Science and Engineering by the Venus Foundation, in 2018, and the Outstanding Researcher Award from SBYTE Educational and Research Foundation, in 2020. She has also been awarded the YUVA Mentor as a Change Maker Award and the Certificate of Appreciation from YUVA Incubated and K. I. T. E. S Education, in November 2021.



RODRIGUEZ RIVERO CRISTIAN received the degree from the Department of Electrical and Electronics Engineering, School of Exact Sciences, Physics, and Natural Sciences, National University of Cordoba, in 2007, and the Ph.D. degree in electronic engineering from the National University of Cordoba, in 2016.

He was a Teaching Assistant with the Department of Electrical and Electronics Engineering, School of Exact Sciences, Physics, and Natural Sciences, National University of Cordoba, until 2013. While pursuing his Ph.D. thesis, in 2013, he joined Aeronautical University Institute as an Adjunct Professor of control systems dynamics. After that, he was a Postdoctoral Fellow in neuroscience with the Centre for Neuroscience, University of California Davis, in 2017, where he focused on a closed-loop phase-locked stimulation project. In 2019, he joined the University of Amsterdam, the Netherlands, as a Lecturer in data science with the Master’s Program for Artificial Intelligence and Information Studies and the Amsterdam Medical Centre, where he received the Basiskwalificatie Onderwijs (BKO)–Dutch University Teaching Qualification. In 2022, he joined the Cardiff School of Technologies, Cardiff Metropolitan University, as a Lecturer in computer science, where he is currently an Academic Associate in U.K. Additionally, he joined Universitat Politècnica de Catalunya as an Associate Professor, in 2024. He has numerous publications in international peer-reviewed conferences and journals. He has also co-founded and been part of the steering committee of the Latin America Society of Computational Intelligence and was the Conference Chair of the 1st Latin America Conference on Computational Intelligence (LA-CCI 2014). Additionally, he has served as the chair, an advisory committee member, and a technical program committee member for various IEEE conferences. He has been a Reviewer for IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS.



R. REENA ROY received the Ph.D. degree, in September 2022, with a thesis titled “An Improved Machine Learning Approach for Diagnosis of Pancreatic Cancer using Multi-Modal Clinical Data” from Anna University, Chennai, India. She is currently an Assistant Professor Senior with Vellore Institute of Technology, Chennai, Tamil Nadu, India. She has more than ten years of experience in teaching and research. She has more than 75 bachelor’s/master’s students who have completed their thesis dissertations under her supervision. She has published more than 20 papers in international journals and conferences and has contributed to three book chapters published by Springer and CRC (Taylor and Francis). Her research interests include applying AI technologies in healthcare, medical image analysis, and artificial intelligence. She received the Best Project Award from Tamil Nadu Science and Technology (TNSCST) for the project titled “Driver Drowsiness Detection System using Image Processing,” in 2019.

...